

# A Mathematical Theory of Value

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## Abstract

We propose that *value* — the quantity that goal-directed agents create, destroy, and exchange — is a lawful structural quantity, in the same category as information, once stripped of its semantic clothing (morality, price, psychology). Following the method of Shannon (1948), we make one ruthless abstraction: value is the rate at which an agent converts a physical resource into goal-progress, relative to a frame fixed by the agent’s goal. From a single scale-invariance axiom we derive a logarithmic measure of value,  $V = \sum_i k_i \ln e_i$ . From the compounding dynamics of a resource that is reinvested we derive a *coding theorem of value*: the rate at which an agent can create value through a perception channel  $Y$  of the world  $X$  is bounded by the mutual information,  $\Delta G \leq I(X; Y)$ , achieved by Bayes-proportional allocation; and realized value decomposes exactly as available potential minus dissipation,  $G = D(q \parallel r) - D(q \parallel p)$ , identifying misalignment with measurable waste. For populations we show value is frame-relative while *price* is frame-independent, that the collective value throughput of a fleet is capped by the world’s entropy  $\sum_a G_a \leq H(X)$ , and that the fleet’s operating point is a Kelly portfolio over agents selected by an emergent price. A dynamical layer gives the equations of motion and an is/ought asymmetry — beliefs have a target the world supplies, goals do not — from which alignment emerges as a control-stability condition with a closed-form residual misalignment. We then test the single-frame laws on live language models: across four local models on a frozen 100-item decision task, perception mutual information tracks realized *capability* rather than parameter count ( $r = 0.99$ ; a cross-family model that is larger yet weaker lands at correspondingly lower  $I$ ), out-of-sample  $\Delta G$  tracks  $I(X; Y)$  ( $r = 0.99$ ), over-confidence is measurable dissipation that shrinks with capability, and diversity-beats-redundancy holds; the fleet-pricing claim is reported with its honest boundary (no volatility-harvesting “demon” on positively correlated agents, but pricing wins under a compute budget). None of the underlying mechanisms is individually new; the contribution is their unification under one substrate-grounded quantity, and the is/ought asymmetry that follows.

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## 1 Introduction

Information theory began with an abstraction that looked like a loss. Shannon (1948) threw away *meaning* and kept only the reduction of uncertainty; from that single move came a measure ( $-\log p$ ), a unit (the bit), and hard limits (channel capacity, the coding theorems). The abstraction was productive precisely because it was ruthless: by refusing to model what a message *meant*, Shannon could state exactly what any communication system can and cannot do.

We attempt the analogous move for *value*. The everyday notion of value is a tangle of morality, market price, and human preference. We throw all three away and keep one structural residue: *value is the rate at which a goal-directed agent converts a scarce physical resource into progress toward its goal*. This is frame-relative — it is defined only relative to a goal, the way information is defined only relative to a prior and energy only relative to a reference frame — but relativity does not disqualify a quantity from being fundamental. It forces the theory to make the frame explicit. A theory of value is to agency what relativity is to motion: the quantity is frame-dependent; the laws relating frames are universal.

The motivation is not only foundational. If, as increasingly seems likely, near-future populations of artificial agents are each driven by an explicit objective, then a theory of value is not philosophy — it is the control theory for agent populations. To route, align, and govern many separated agents

we need value to be a *measured* quantity with conservation laws and limits, not a verdict. This paper builds that quantity and tests it on real agents.

**Contributions and honesty.** We claim none of the underlying mechanisms as individually novel. The logarithm is Bernoulli’s and Kelly’s; the growth-rate-as-information-rate is Kelly’s; the price layer is Arrow–Debreu’s; the information geometry is Čencov’s and Amari’s. Our contribution is fourfold and we state it plainly (§11): (i) the *unification* of these accounts under one substrate-grounded quantity; (ii) the *capacity theorem* and *fleet ceiling* that the unification makes visible; (iii) the *is/ought asymmetry* as a structural property of agent dynamics; and (iv) an *empirical test* on live models. Throughout, each result is followed by the assumptions that make it bite, in Shannon’s spirit of naming the model.

**Plan.** Part I (statics) derives the measure, the capacity theorem, and the Second Law for a single agent. Part II (multi-agent) develops cross-frame price and the fleet capacity region. Part III (dynamics) gives the equations of motion and alignment-as-stability. Part IV (evidence) reports the synthetic and real-agent tests. Appendices collect the two proofs.

## Part I

# Statics: the measure, the limit, the Second Law

## 2 The measure: a logarithmic law of value

**Setup.** An agent holds a divisible resource of total size  $E > 0$  (free energy, compute, budget, attention — any fungible substrate of action) and allocates it across  $K$  goal-relevant channels in amounts  $e = (e_1, \dots, e_K)$ ,  $\sum_i e_i = E$ . Each channel  $i$  carries a *goal weight*  $k_i \geq 0$  measuring how much progress toward the goal a unit of effective work on  $i$  yields. Let  $V(e)$  be the value the allocation realizes. We ask what functional form  $V$  must take.

**Axiom 1** (Diminishing returns).  $V$  is increasing and strictly concave in each  $e_i$ : the first unit of resource on a channel is worth more than the thousandth.

**Axiom 2** (Additivity across independent channels). For channels whose contributions do not interact,  $V(e) = \sum_i v(e_i; k_i)$ .

**Axiom 3** (Scale invariance). Rescaling the resource unit ( $e_i \mapsto \lambda e_i$ ) changes total value only by an additive constant independent of the allocation:  $v(\lambda e_i; k_i) = v(e_i; k_i) + c(\lambda, k_i)$ .

Axiom 3 is the load-bearing one: it says value has no privileged zero of *scale*, exactly as information has no privileged base of logarithm. It is a Cauchy functional equation in disguise.

**Theorem 1** (Logarithmic Value Law). *Under Axioms 1–3 with mild regularity, the value measure is, up to choice of unit,*

$$V(e) = \sum_{i=1}^K k_i \ln e_i \tag{1}$$

and the value-maximizing allocation of a fixed budget  $E$  is proportional to the goal weights,  $e_i^* = E \hat{k}_i$  with  $\hat{k}_i = k_i / \sum_j k_j$ , giving

$$V^* = K \ln E - K H(\hat{k}), \quad H(\hat{k}) = - \sum_i \hat{k}_i \ln \hat{k}_i. \quad (2)$$

The proof (Appendix A) solves the Cauchy equation forced by Axiom 3. Two features deserve note. First, Shannon's entropy  $H$  reappears *unbidden* as a penalty: a goal spread thinly over many channels (high  $H(\hat{k})$ ) realizes less value than a focused one — *focus is worth  $K H(\hat{k})$  nats of value*. Second, the logarithm here is forced by diminishing returns; §3 forces it a *second* time from compounding. A quantity whose form is over-determined by two independent routes is the signature of something fundamental, not a modeling convenience.

### 3 The capacity theorem: value-throughput is bounded by information-throughput

**The repeated game.** Now let value compound: value realized this round becomes resource next round (an agent that achieves its goal earns more budget, trust, or compute). Each round  $t$  the world settles into a state  $s \sim q$ ; the agent splits its budget into fractions  $b = (b_1, \dots, b_n)$ ,  $b_i \geq 0$ ,  $\sum_i b_i = 1$ ; and channel  $i$  returns a multiplier  $o_i(s) \geq 0$  per unit committed. The budget updates multiplicatively,  $E_{t+1} = E_t \sum_i b_i o_i(s_t)$ , so  $\log E_T = \log E_0 + \sum_t \log(\sum_i b_i o_i(s_t))$  and the long-run *value growth rate* is

$$G(b) = \mathbb{E}_{s \sim q} \left[ \log \sum_i b_i o_i(s) \right]. \quad (3)$$

That the time-average growth of a multiplicative process is the *expected log* of its multiplier is the ergodicity argument of Peters (2019); it forces the logarithm again, now from dynamics rather than psychology.  $G$  is concave in  $b$ , so maximizing it is a well-posed convex program.

**The horse race.** Take the canonical structure in which states are the channels, committing to channel  $i$  pays only in state  $i$ , and returns are quoted against a *reference belief*  $r$  over states,  $o_i = 1/r_i$ . Then  $G(b) = \sum_s q_s \log(b_s/r_s)$ , maximized (Gibbs) by betting one's model,  $b_s = p_s$ . With a correct model  $p = q$ ,

$$G^* = \sum_s q_s \log \frac{q_s}{r_s} = D(q \| r). \quad (4)$$

*Maximal value growth equals the divergence of the agent's correct model from the baseline.* An agent whose model is the baseline ( $p = r$ ) grows value at rate zero: value creation is informational edge over the prevailing expectation, nothing more.

**Side information.** Equip the agent with a perception channel: a signal  $Y$  correlated with the world-state  $X$ , observed before allocating. Optimal play bets the posterior  $b_s = p(s | y)$ , with growth  $G_Y = \sum_{x,y} p(x,y) \log \frac{p(x|y)}{r(x)}$ . The gain over acting on the prior alone is exactly mutual information:

**Theorem 2** (Coding Theorem of Value). *The incremental rate at which an agent can compound value by perceiving the world through a channel  $Y$  is at most the mutual information between world-state and perception,*

$$\boxed{\Delta G = G_Y - G_0 = I(X; Y)}, \quad (5)$$

*and this bound is achieved by Bayes-proportional allocation.* (Proof: Appendix B.)

This has the full structure of a Shannon coding theorem. *Achievability*: betting the posterior attains  $\Delta G = I(X; Y)$ . *Converse*: any  $b \neq p(\cdot | y)$  loses exactly  $D(p(\cdot | y) \| b)$  per round (Gibbs), so  $I(X; Y)$  is an upper bound. An agent cannot create value faster than it can perceive the world; *value-throughput is bounded by information-throughput*. Perception capacity is the hard ceiling on value-generation rate — the value analog of channel capacity.

## 4 The Second Law of Value

Drop the assumption that the agent’s model is correct. Let  $q$  be reality,  $p$  the agent’s model,  $r$  the baseline; the agent bets  $b = p$  but the world is drawn from  $q$ . Then

$$G_{\text{actual}} = \sum_s q_s \log \frac{p_s}{r_s} = \underbrace{D(q \| r)}_{\text{available potential}} - \underbrace{D(q \| p)}_{\text{dissipation (model error)}}. \quad (6)$$

Realized value is available potential minus dissipation from misalignment, every term in nats. Three consequences. (i) *Misalignment is measurable dissipation*: confident error ( $D(q \| p) > D(q \| r)$ ) drives growth negative — a wrong, certain agent actively destroys value, while calibrated humility is value-preserving. (ii) *Learning is value-recovery*:  $D(q \| p)$  is exactly the cross-entropy excess that predictive training minimizes, so the machine-learning objective *is* the minimization of value dissipation. (iii) *Alignment is a value law*: reducing  $D(q \| p)$  is mechanically identical to reducing wasted value. We test (6) on live models in §10 (R2), where the weakest model’s over-confidence indeed drives realized growth negative.

## Part II

# Multi-agent: price and the fleet

## 5 Cross-frame value and the frame-independence of price

A single agent’s value lives in its own frame  $(q, p, r, k)$ . Two agents with different goals have values that are *not* cardinally comparable — the interpersonal-comparison impossibility of Arrow (1951), which we concede rather than circumvent. Yet they can still coordinate, exactly as economies coordinate non-comparable utilities: through a *price* on the shared resource.

**Shadow price.** For an agent with goal mass  $K = \sum_i k_i$  and budget  $E$ , the marginal value of resource (the Lagrange multiplier on  $\sum_i e_i = E$  in Theorem 1) is the *shadow price*  $\lambda = K/E$ . Two agents trade resource until their shadow prices equalize; the common  $\lambda$  is a scalar that lives in *neither* agent’s value frame. Hence:

*Value is frame-relative; price is frame-independent.* Value is a vector in an agent’s goal basis; price is the one scalar on which separated frames agree.

**The invariant.** The transformation between two agents’ value frames is a change of coordinates on the belief simplex. The unique (up to scale) Riemannian metric invariant under such sufficient-statistic reparametrizations is the Fisher–Rao metric (Čencov, 1982; Amari, 2016). It is the cross-frame invariant of the theory: distances in belief that all agents agree on, whatever their goals. Alignment between two goals is the cosine  $\cos \theta_{ab} = \langle \hat{k}_a, \hat{k}_b \rangle$ ; its sign sets whether their interaction is positive- or negative-sum.

## 6 The fleet capacity region

Consider  $m$  agents acting on one world  $X$  and drawing on one resource pool.

**The ceiling.** Each agent’s growth is bounded by its own perception (Theorem 2),  $G_a \leq I(X; Y_a)$ . Jointly, the data-processing inequality gives the sum-rate bound

$$\boxed{\sum_{a \in S} G_a \leq I(X; Y_S) \leq H(X)} \quad \text{for any subset } S, \quad (7)$$

the collective value throughput of a fleet is capped by the entropy of the world it acts on. Headcount past perception-saturation buys nothing. *Perception diversity* (agents that perceive different slices of  $X$ ) lifts  $I(X; Y_S)$  toward  $H(X)$ ; *redundancy* (agents perceiving the same slice) adds exactly zero. We confirm both on live models in §10 (R4).

**The operating point.** Under multiplicative dynamics the shared pool multiplies each round by  $\sum_a w_a R_a$  for resource weights  $w_a$ , so the fleet is a *Kelly portfolio over agents*: for imperfectly-correlated agents, spreading resource and rebalancing raises the time-average growth (the volatility-harvesting effect of Kelly, 1956; Cover, 1991). The weights that achieve the optimum are selected by the resource *price*  $\pi$ : price moves resource to high-shadow-price agents until  $\lambda$  equalizes. Thus

the *alignment matrix*  $M = [\cos \theta_{ab}]$  shapes the achievable region (a cooperation dividend where  $\cos \theta > 0$ , a conflict tax where  $\cos \theta < 0$ ); the *price* selects the operating point on it.

This separates governance into two acts: *shaping* the region (alignment and perception design) and *choosing* the point on it (pricing). The pricing claim is the one we test most carefully (§10, R5), and where the demon’s precondition — genuine diversity — turns out to matter.

## Part III

# Dynamics: motion and alignment

## 7 The equations of motion and the is/ought asymmetry

The statics fix equilibria; the dynamics say how they are approached. Three objects evolve. *Beliefs*  $p$  flow toward the true law  $q$  by a natural-gradient (Fisher) descent on log-loss — the realized regret of this flow equals the cumulative dissipation  $\sum_t D(q \| p_t)$  of (6), so *learning is value-recovery* dynamically as well as statically. *Prices*  $\pi$  flow toward market-clearing by tâtonnement (dual ascent on the resource constraint). *Goals*  $k$  flow under two forces: *control* (a principal pulling  $k_a$  toward a target  $k^*$ ) and *selection* (compounding reweights resource toward higher-growth goals).

The organizing fact is an asymmetry. Beliefs and prices each flow toward a target *the world supplies*: reality provides  $q$ , the resource constraint provides market-clearing. Goals have *no* target the world supplies — reality contains no fact about what ought to be valued. So beliefs and prices are *learnable* (gradient flows toward a given target); goals are only *controllable* or *selectable*. This is Hume’s is/ought gap recovered as a structural property of the dynamics, and §8 shows it is the mathematical shape of the alignment problem.

## 8 Alignment as a stability condition

Let agent  $a$  have goal  $k_a$  and resource share  $w_a$ , with fleet effective goal  $\bar{k} = \sum_a w_a k_a$ . Control pulls each goal toward target:  $\dot{k}_a|_{\text{ctrl}} = -\gamma(k_a - k^*)$ . Selection follows the replicator  $\dot{w}_a = w_a(G_a - \bar{G})$ . Summarize the environment’s reward landscape over goals to first order by its growth gradient  $g := \nabla_k G$  — the direction in goal-space along which deviating from  $k^*$  *increases* resource capture.

**Theorem 3** (Coupled-flow alignment). *By Price’s equation (Price, 1970), the selection drift of the mean is the trait–fitness covariance; with  $G_a \approx \bar{G} + g^\top(k_a - \bar{k})$  it equals  $Vg$ , where  $V = \text{Cov}_a(k_a)$  is the goal-dispersion matrix. Hence the effective goal obeys*

$$\dot{\bar{k}} = -\gamma(\bar{k} - k^*) + Vg, \quad (8)$$

with fixed point and residual misalignment

$$\boxed{\bar{k}^* = k^* + \gamma^{-1}Vg, \quad \|\bar{k}^* - k^*\| = \gamma^{-1}\|Vg\|,} \quad (9)$$

and the aligned fixed point is locally stable iff  $\gamma > \lambda_{\max}(\partial(Vg)/\partial\bar{k})$ .

The fleet does not in general settle at the target; it settles a distance  $\|Vg\|/\gamma$  away — goal-dispersion times reward-pull over control gain. Perfect alignment holds iff  $Vg = 0$ : either the environment rewards exactly the target ( $g = 0$ ), or there is no diversity to select among ( $V = 0$ ), or control is infinite. The governance reading is sharp:

*The cheap half of alignment is incentive design, not control.* Driving  $g \rightarrow 0$  (aligning what *pays* with what is *wanted*) removes the residual for *any*  $\gamma$  and *any*  $V$ , preserving the diversity that lifts the fleet ceiling (7); raising  $\gamma$  (brute-force oversight) merely opposes a drift it never removes. Spend first on  $g$ .

This is the actionable form of the is/ought asymmetry: because goals have no world-given target, they are governed by what pays (selection, via  $g$ ) and what is imposed (control, via  $\gamma$ ).

## Part IV

# Evidence

## 9 Synthetic validation

We first check that each closed-form prediction is reproduced by a Monte-Carlo world built to the theory’s assumptions. All five families pass to numerical tolerance (20/20 checks): E1,  $\Delta G = I(X; Y)$  to  $\sim 10^{-3}$  nats; E2, the Second-Law decomposition (6) exactly, including value going negative under confident error; E3, the fleet ceiling (7) with diversity lifting it and redundancy adding zero; E4, a Kelly-priced fleet beating ad-hoc allocation, including a “Shannon’s demon” built from anti-correlated agents that individually do not grow yet collectively do; E5, cumulative dissipation equal to regret, with a drifting world flooring dissipation at a positive value (a dynamical Second Law). These confirm the mathematics is self-consistent and correctly derived. They are, however, circular by construction: the worlds are drawn from the same distributions the formulas assume. The decisive test is on real agents.

## 10 Real agents

**Instantiation.** We take a frozen, pre-registered 100-item decision task in which each item has a ground-truth correct action drawn from  $K = 7$  classes, and an agent’s output *is* an action. This realizes the perception-then-act structure of Theorem 2 directly: the world-state  $X$  is the correct action, the perception  $Y_a$  is the action model  $a$  chooses. We set  $r = q$  (the action marginal), so a no-signal agent grows value at rate 0. From each model’s run we form the confusion matrix  $C_a[x, y] = \#(\text{gold} = x, \text{chosen} = y)$  and compute every quantity in nats with the same primitives used in §9:  $I(X; Y_a)$  from the joint; the calibrated posterior  $p_a(x | y) = C_a[:, y] / \sum_x C_a[:, y]$ ; and realized growth  $\Delta G_a = \text{mean} \ln(p_a(x | y_a) / r(x))$ . The agents are four local models spanning a range of realized capability — three of one family (1.5B, 3B, 7B) and a cross-family model (an 8B that is, on this task, *less* capable than the 7B) — greedy decoding. Posteriors are calibrated on a 48-item fit split and scored on a 52-item holdout; in-sample, the oracle posterior with  $r = q$  makes  $\Delta G = I$  an arithmetic identity (used only to confirm the units), so the empirical content is the capability tracking, the out-of-sample tracking, and the fleet result.

**R1: the bridge holds, and  $I$  tracks capability, not size.** Mutual information tracks realized capability (tool-accuracy) at Pearson  $r = 0.99$ , and out-of-sample value-growth tracks  $I$  at  $r = 0.99$  (Table 1). The decisive datum is the cross-family model D: it is *larger* than C (8B vs 7B) yet *less* capable (0.86 vs 0.96), and its  $I$  lands accordingly below C, near B. So the claim is not “scale buys value” but the sharper one: value-throughput is information-throughput, and information-throughput is set by what the model can actually perceive.

| model            | params | tool-acc | $I(X; Y)$ | $\Delta G_{\text{in}}$ | $\Delta G_{\text{hold}}$ |
|------------------|--------|----------|-----------|------------------------|--------------------------|
| A                | 1.5B   | 0.79     | 1.277     | 1.277                  | 0.915                    |
| B                | 3B     | 0.87     | 1.561     | 1.561                  | 1.242                    |
| C                | 7B     | 0.96     | 1.779     | 1.779                  | 1.423                    |
| D (cross-family) | 8B     | 0.86     | 1.513     | 1.513                  | 1.109                    |

Table 1: R1 (nats).  $I(X; Y)$  tracks realized capability, not parameter count ( $r = 0.99$  vs accuracy);  $\Delta G_{\text{hold}}$  tracks  $I$  ( $r = 0.99$ ). Model D is larger than C yet weaker, and its  $I$  lands below C accordingly. World entropy  $H(X) = 1.92$  nats: C captures 93%, D (though larger) only 79%, A 66%.

**R2: over-confidence is dissipation, shrinking with capability.** Reading the same point-predictions with a calibrated versus an over-confident posterior realizes different value; the gap is dissipation (Table 2). It is large for the weak models and shrinks with capability (not size: the larger-but-weaker D dissipates more than C) — a quantitative statement that less-capable agents must be more humble. For models A and D, confident error drives realized growth *negative*, exactly the Second Law’s prediction (6).

**R3: value per joule.** Mutual information *per second of compute* falls across the family ( $0.74 \rightarrow 0.38 \rightarrow 0.30$  nats/s for A,B,C), and the cross-family D is worst (0.15 nats/s: slowest and lower  $I$ ). The weakest model is the most efficient perceiver per unit compute — the value-theoretic reason a cheap reflex is worth running on a narrow task. (This is the  $I/\text{compute}$  curve across heterogeneous models, not a within-model prompt ablation; see §13.)



| model    | tool-acc | $G_{\text{cal}}$ | $G_{\text{over}}$ | dissipated |
|----------|----------|------------------|-------------------|------------|
| A (1.5B) | 0.79     | +0.915           | -3.255            | 4.17       |
| B (3B)   | 0.87     | +1.242           | -0.863            | 2.11       |
| C (7B)   | 0.96     | +1.423           | +0.731            | 0.69       |
| D (8B)   | 0.86     | +1.109           | -1.660            | 2.77       |

Table 2: R2 (nats). Dissipation from false certainty falls as *capability* rises (not size: the larger but weaker D dissipates more than C); for the least-capable models over-confidence destroys value outright.

**R4: diversity beats redundancy.** Two different models jointly perceive more of  $H(X)$  than either alone; an identical greedy re-run adds exactly zero. The highest joint information comes from the strongest diverse pair (B+C,  $I = 1.869$  nats, +0.090 over C, closing most of the gap to  $H(X) = 1.921$ ), while the largest *lifts* come from pairing differently-wrong weak models (A+B and B+D each add  $\sim 0.2$  nats). This is the fleet-ceiling prediction (7) on real agents.

**R5: pricing beats pooling, but only where there is diversity to price (reported honestly).** On raw out-of-sample growth, the Kelly/price fleet (+1.328) beats the equal-weight ensemble (+1.315) but *does not beat the single best model* (+1.423). There is no Shannon’s demon here, and the theory says why: four models on the *same* task are positively-correlated agents (R4 quantifies the best residual diversity at +0.09 nats), so Kelly rebalancing has no anti-correlated volatility to harvest and the best agent dominates. This is the honest negative the synthetic E4 demon does not reproduce, because E4 was constructed with anti-correlated agents.

The negative is confined to the cost-blind axis. Value *per unit compute* inverts the ranking: model A yields 0.531 nats/s against C’s 0.243 (and the slow D only 0.111), and a budget-aware price  $\propto I_a/\text{cost}_a$  (the shadow price  $\lambda = K/E$  of §5) achieves 0.328 nats/s, beating the best single model’s density. Pricing therefore pays exactly where §6 says it should: as the lever that chooses the operating point under a resource constraint, not as a free lunch that beats the best agent when compute is unlimited. The demon needs *perception diversity* — different slices of  $H(X)$ , i.e. specialists rather than generalists of varying strength — which a set of same-task generalists lacks by construction. We mark this as a falsifiable boundary, not a defeat.

## 11 Related work

Every component of this theory exists in some field; the contribution is the unification, the substrate grounding, and the is/ought asymmetry. We state this explicitly and concede each component to its source.

*Expected utility and diminishing value.* The axiomatic treatment of preference is von Neumann and Morgenstern (1944) and Savage (1954); our logarithmic measure (§2) coincides with Bernoulli (1738) and with unit-coefficient constant-relative-risk-aversion utility, and the diminishing-returns law it encodes is Weber–Fechner. We do not claim the form as novel; our departure is to *derive* it from scale invariance on a physical resource, to ground it in a substrate (free energy) supplying a unit and a cross-frame exchange rate that expected utility lacks, and to show it is forced a second time by compounding (Peters, 2019).

*Information rate and log-optimal growth.* The closest prior art, which we credit most carefully, is Kelly (1956): the exponential growth rate of wealth is an information rate, optimized by betting

one’s beliefs; Breiman (1961) proved asymptotic optimality and Cover (1991) developed universal portfolios. Our capacity theorem (§3) is generalized Kelly, and the fleet operating point (§6) is a Kelly–Cover portfolio over agents. What we add is the reinterpretation of wealth as free energy (a physical rather than monetary substrate), the cross-frame price layer, the dynamics/alignment layer, and the claim that Kelly’s theorem is the single-agent monetary *special case* of a general law of value.

*Reinforcement learning.* “Value function” is precisely defined in RL (Bellman, 1957; Sutton and Barto, 2018) as expected cumulative reward; the relationship is complementary, not competitive. RL takes reward as given and maximizes it; our goal weights are the analog of reward, but we problematize where reward comes from and how it drifts (§7–8). Our theory sits beneath RL — a candidate account of what reward *is*.

*Thermodynamics of computation and the free-energy principle.* That information has thermodynamic cost is Landauer (1961) and Bennett (1982); Jaynes (1957) recast statistical mechanics as inference; England (2013) studied dissipation-driven adaptation; Friston (2010) casts agents as free-energy minimizers. Our belief dynamics (§7) is free-energy-principle-like and shares the information geometry (Čencov, 1982; Amari, 2016). The distinction is precise: the free-energy principle is a theory of the perception/belief half — the “is” — whereas our value layer is the goal/value half — the “ought” — plus a multi-agent price economics the FEP does not contain. The seam between them is the is/ought asymmetry.

*General equilibrium and social choice.* Our price layer (§5) is Arrow and Debreu (1954); Debreu (1959) and its companion impossibility (Arrow, 1951). We use these rather than extend them: we concede that cardinal cross-agent comparison is impossible and route around it via an emergent price. The contribution is the synthesis with the information-theoretic value layer under one substrate, and the application to artificial-agent populations.

*AI alignment.* Instrumental convergence (Omohundro, 2008; Bostrom, 2014) we derive from the compounding dynamics of value (§7): goal-directions correlated with resource capture are selected regardless of terminal content. We recast alignment as a dynamical stability condition (§8), which to our knowledge is a new framing.

## 12 Discussion: governing a population of agents

The results compose into a control theory for populations of separated agents. *Per agent*, value-generation rate is capped by the perception mutual information  $I(X; Y_a)$  (Theorem 2); supplying an agent the goal-relevant bits cheaply — raising effective  $I$  per unit compute — lifts its value ceiling without raising its deliberation cost, which is the formal reason a cheap reflex can match expensive deliberation on a narrow task (R3 measures this curve directly). *Across agents*, values are not cardinally comparable, so coordination must run through an emergent price on shared resource (§5), never a god’s-eye utility sum. *Fleet design* is then two levers: cultivate perception diversity to lift the entropy ceiling (7) (R4), and price resource to the best operating point (R5) — with the caveat, learned empirically, that pricing earns its keep only where genuine diversity exists. *Alignment*, finally, is incentive design before oversight (Theorem 3): spend first on making what pays equal what is wanted.

## 13 Limits

We name the assumptions that bound each claim. The capacity theorem assumes compounding (reinvested value) and is cleanest in the horse-race structure; general returns give a concave program

without the closed-form  $D(q \parallel r)$ . The fleet ceiling’s exact achievable region is an open problem of network-information-theory difficulty. The cross-frame layer concedes that cardinal interpersonal comparison is non-canonical; we do not solve it. On the empirical side: the in-sample  $\Delta G = I$  is an arithmetic identity, not evidence — only the capability tracking, the out-of-sample tracking, and R5 are empirical; R3 is an  $I/\text{compute}$  curve across model scale, not a within-model prompt ablation; the calibration in R2 is constructed from the confusion matrix, not a probability the model reported, so the dissipation measured is that of a stated belief over those predictions; and the whole test is one task on one corpus — evidence that the bridge holds there, not a universal law. Degenerate runs in which a model emits no action (a serving/plumbing failure, not a decision) are detected and excluded rather than reported as low- $I$  capability. The alignment theorem is mean-field with a linearized landscape and a quasi-static dispersion  $V$ ; a fully coupled treatment, and an endogenous reward gradient  $g$  in a closed multi-agent world, remain open.

## 14 Conclusion

Stripped of morality, price, and psychology, value is a structural quantity with a measure ( $\sum_i k_i \ln e_i$ ), a capacity limit ( $\Delta G = I(X; Y)$ ), a Second Law ( $G = D(q \parallel r) - D(q \parallel p)$ ), a frame-independent price, a fleet entropy ceiling, and an alignment-stability law. Shannon’s own quantities — entropy, divergence, mutual information, the Fisher metric — reappear unbidden throughout, which is the strongest internal evidence that the abstraction is real rather than decorative. On live language models the single-frame laws hold to high precision, and the one fleet-level claim with a genuine precondition is reported with its boundary intact. The quantity is frame-relative; the laws relating frames are universal; and they are now testable on the artificial agents the theory was built to govern.

## A Proof of the Logarithmic Value Law (Theorem 1)

By Axiom 2 it suffices to find the per-channel  $v(e; k)$ . Axiom 3 states  $v(\lambda e; k) = v(e; k) + c(\lambda, k)$  for all  $\lambda, e > 0$ . Fix  $k$  and write  $f(e) := v(e; k)$ . Then  $f(\lambda e) - f(e) = c(\lambda)$  is independent of  $e$ . Differentiating in  $\lambda$  at  $\lambda = 1$ ,  $e f'(e) = c'(1) =: k$ , a separable ODE with solution  $f(e) = k \ln e + \text{const}$ . Concavity (Axiom 1) fixes the sign  $k \geq 0$  and rules out the additive constant’s dependence on  $e$ . Summing over channels gives  $V(e) = \sum_i k_i \ln e_i$ . Maximizing under  $\sum_i e_i = E$  with multiplier  $\lambda$ :  $\partial_{e_i}(V - \lambda \sum_j e_j) = k_i/e_i - \lambda = 0 \Rightarrow e_i = k_i/\lambda$ , and  $\sum_i e_i = E$  gives  $\lambda = K/E$  (the shadow price) and  $e_i^* = E \hat{k}_i$ . Substituting,  $V^* = \sum_i k_i \ln(E \hat{k}_i) = K \ln E + \sum_i k_i \ln \hat{k}_i = K \ln E - K H(\hat{k})$  after normalizing  $k_i = K \hat{k}_i$ .  $\square$

## B Proof of the Coding Theorem of Value (Theorem 2)

*Achievability.* With side information  $Y = y$  the optimal bet is the posterior  $b_s = p(s \mid y)$  (Gibbs applied conditionally), giving conditional growth  $\sum_s p(s \mid y) \log \frac{p(s \mid y)}{r(s)}$ . Averaging over  $y$ ,

$$G_Y = \sum_y p(y) \sum_s p(s \mid y) \log \frac{p(s \mid y)}{r(s)} = \sum_{x,y} p(x, y) \log \frac{p(x \mid y)}{r(x)}.$$

Without side information the optimal bet is the prior  $b_s = p(s)$ , giving  $G_0 = \sum_x p(x) \log \frac{p(x)}{r(x)} = D(p_X \parallel r)$ . Subtracting,

$$\Delta G = G_Y - G_0 = \sum_{x,y} p(x,y) \log \frac{p(x|y)}{p(x)} = I(X;Y).$$

*Converse.* For any allocation  $b(\cdot | y) \neq p(\cdot | y)$ , the conditional growth loss is  $\sum_s p(s | y) \log \frac{p(s|y)}{b(s|y)} = D(p(\cdot | y) \parallel b(\cdot | y)) \geq 0$  by Gibbs, with equality iff  $b = p(\cdot | y)$ . Hence no strategy exceeds  $G_0 + I(X;Y)$ , so  $I(X;Y)$  is both achieved and an upper bound on  $\Delta G$ .  $\square$

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